

An Analytical Framework for Error and Hallucination in Deployed Generative AI Systems

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Abstract

Evaluation of errors in a deployed generative AI system is difficult when the system combines a trained model with designated sources, retrieved information, tools, prior interactions, and system controls. Under these conditions, an error in the response delivered to a user may not originate in the trained model. The information used to produce the response may also differ from the information that should govern its evaluation. To address these problems, this paper develops an authority-first evaluation framework. The framework first identifies the governing reference for each assessable claim and then classifies the relationship between the claim and that reference as supported, contradicted, unsupported, or unresolved. It treats hallucination and task-obligation fidelity error as related error classifications that may overlap. Hallucination concerns a representational claim that is contradicted by, or lacks support required from, its governing reference; task-obligation fidelity error concerns failure to satisfy an applicable requirement. The framework further separates identification of an error from its localization and causal attribution, and separates occurrence of the error from its consequences. Applications to temporal change, continuing-project state, retrieval-augmented generation, and error propagation illustrate why evaluation should begin with authority rather than with an assumed source of failure. This sequence may support more consistent evaluation and better-targeted controls, while keeping the cause of a response error unresolved when the available evidence does not support a firmer conclusion.

Key words: AI governance; AI evaluation; model error; hallucination; factuality; faithfulness; governing reference; task-obligation fidelity; project-state fidelity; RAG

A. Introduction

Generative AI applications increasingly operate as deployed systems rather than as stand-alone models. A deployed system may combine a trained model with an organization’s documents, retrieved information, software tools, prior interactions, and technical or procedural controls. When work continues across time, the system’s response to a current request may also depend on decisions and records created during earlier work.

This broader system setting creates two evaluation problems. The first arises when the information used to produce a response is not the information that should govern its evaluation. A system may, for example, receive an outdated document when a current rule governs, or rely on general knowledge when the task requires fidelity to a designated source. Evaluation of the response would then be misleading unless the applicable reference is identified before the response is judged.

The second problem arises because an observed error does not, by itself, explain how the error was produced. A false claim may reflect a limitation of the trained model, omission of relevant information during context construction, failure during generation to use supplied information, or transformation of the response before delivery. Moreover, the same production problem may lead to different outcomes. Failed retrieval may be followed by abstention in one case and fabrication in another. Identification of an error, localization of its possible source, and attribution of its cause should therefore be treated as separate questions.

Calling a response error a hallucination does not resolve either problem. Existing definitions of hallucination emphasize, in varying degrees, factual falsehood, lack of evidentiary support, or inconsistency with supplied material ([1]–[5]). Research also distinguishes a contradiction from an unverifiable addition, and faithfulness to a source from factual agreement with the external world ([6], [7]). These distinctions are useful once the reference applicable to a claim has been established. They do not, however, determine which reference should govern the claim.

This paper begins with that question. We use the term *authority-first evaluation* for a sequence that begins by selecting the governing reference and then proceeds to discrepancy classification, localization and causal attribution, consequence assessment, and governance response. The sequence applies after a response has been delivered. It does not assume that every deficiency originated in the trained model, and it does not require a causal conclusion when the available evidence establishes only that a discrepancy occurred.

The framework also separates hallucination from task-obligation fidelity. Task-obligation fidelity means conformity with requirements established by a governing source, instruction, contextual record, or authorized project state. By authorized project state, we mean the accepted decisions, requirements, and statuses governing continuing work at the time of the task. An approved requirements document may therefore form part of the authorized state, whereas an unapproved proposal or a superseded draft does not. A task-obligation fidelity error may be entirely behavioral, such as exceeding a word limit. It may also overlap with hallucination if the response falsely represents an instruction, source, record, or project decision.

The analysis is conceptual rather than empirical. Its intended contribution is a structured way to ask four questions: What governs a claim? What discrepancy occurred? What can the available evidence establish about how the discrepancy was produced? What response is appropriate? Section B describes how a deployed system produces a delivered response. Section C develops governing-reference selection and discrepancy evaluation. Section D distinguishes hallucination from task-obligation fidelity, and Section E separates error-source localization from causal attribution. Section F applies the framework to time, continuing-project state, and retrieval-augmented generation. Section G considers consequences, controls, limitations, and conclusions.

B. Response Production in a Deployed System

To make the production setting explicit, consider a deployed system at interaction time t . The system receives the current input x_t , has access to prior interaction history $\mathcal{H}_{t-1}$, and may draw on external resources E_t , including documents, databases, tools, and current observations. A context-construction

process R^L , operating under an effective constraint L , retrieves, selects, compresses, orders, or labels information to produce the operative context:

$$C_t = R^L(x_t, \mathcal{H}_{t-1}, E_t). \quad (1)$$

We use operative context to mean the supplementary information C_t actually presented for generation. The operative context is not the same as all information accessible to the system. Relevant material may remain outside the operative context because the material was not retrieved, was displaced by other material, or lost an important qualification during compression. Nor does inclusion ensure effective use: model performance may depend on where relevant information appears in a long context ([8]). The current input x_t has two roles in this formulation. It guides construction of the operative context and is also supplied directly to the generation process as the user's current request.

After the operative context has been constructed, let $M(t_0)$ denote the trained model used at response time t , where t_0 is used as a simplified marker for the latest information available during training. Conditional on the current input and operative context, the deployed generation process produces a candidate response:

$$\tilde{y}_t \sim P_M(\cdot | x_t, C_t). \quad (2)$$

Equation (2) summarizes the generation stage; it is not intended to specify a decoding algorithm. The model, current input, and operative context determine conditional predictions, while decoding and successive selections realize one response path. The interval between t_0 and t means that information learned by the model may be outdated. That interval, however, does not by itself determine the delivered response. Current information, decoding, tools, or later controls may correct, transmit, or amplify the resulting susceptibility.

Generation need not be the final stage. Before delivery, a post-generation process may verify, qualify, redact, rewrite, refuse, or block the candidate response. Let $\mathcal{E}$ denote the applicable rules and controls, and let $I_{\mathcal{E}}$ denote the process that implements them:

$$y_t = I_{\mathcal{E}}(\tilde{y}_t, x_t, C_t). \quad (3)$$

The delivered response y_t may therefore differ from the candidate response. This distinction becomes material if a control removes a necessary qualification, inserts an explanation, or prevents an erroneous claim from reaching the user.

Taken together, Equations (1)–(3) distinguish four objects relevant to evaluation: information potentially accessible to the system, the operative context actually supplied, the candidate response generated, and the response ultimately delivered. The equations do not constitute a causal model of response error. Their main purpose is to show why a deficiency in the delivered response should not automatically be attributed to the trained model.

C. Governing References and Response Evaluation

Given a delivered response to a task, evaluation first requires identification of the claims that can be assessed. Following the claim-level approach used in factuality evaluation ([9]), we call each such claim an *assertion*: an explicit or reasonably implied claim whose relation to an applicable reference can be evaluated. A response may contain several assertions, and different assertions in the same response may require different governing references.

Evaluation of an assertion requires selection of the applicable reference. This selection depends on a task specification q_t and an authority policy Π_t . The task specification states what the system is asked to do. The authority policy assigns priority among references when more than one reference may appear relevant to the task. For a source-faithful summary, for example, the policy may specify that the designated source governs even if the model’s external knowledge conflicts with that source. Establishing the policy before examining a preferred answer reduces the risk of selecting a convenient reference after the fact.

More formally, let a_{tj} denote assertion j in the delivered response, and let k_{tj} denote the assertion’s role. If ρ is the reference-selection rule, the governing reference is:

$$r_{tj}^* = \rho(q_t, k_{tj}; \Pi_t). \quad (4)$$

The governing reference is therefore an evaluative standard; it need not be information that the model received. An assertion may agree with the operative context and still conflict with the governing reference. Conversely, it may be correct as a statement about the external world and still fail to represent a designated source faithfully. Table 1 summarizes the principal assertion roles and the types of reference that may govern each role.

Table 1. Assertion roles and governing references

Assertion role	Potential governing reference	Illustration
External factual claim	Current verifiable information at the time relevant to the task	Who currently holds an office; a current price or rule
Designated-source claim	Designated source, including its provenance and qualifications	What a supplied paper states
Claim about an applicable instruction	Applicable instruction	Whether the task required a specified scope, format, or prohibited transformation
Contextual claim	Authoritative contextual record	What was established earlier in the operative exchange
Project-state claim	Authorized project state	Whether a proposal was accepted, revised, or superseded

Once the governing reference has been selected, the assertion can be compared with that reference. Let v denote the comparison rule and V_{tj} the resulting classification:

$$V_{tj} = v(a_{tj}, r_{tj}^*),$$

$$V_{tj} \in \{\text{supported}, \text{contradicted}, \text{unsupported}, \text{unresolved}\}. \quad (5)$$

We call an assertion *supported* when the governing reference supplies the support required by the task, and *contradicted* when the reference conflicts with the assertion. An assertion is *unsupported* when the task requires support that the governing reference does not supply. The relation is *unresolved* when the evaluator cannot determine whether the assertion is supported, contradicted, or unsupported. An unresolved classification does not establish that the assertion is supported and should not be treated as a passing result.

Not every feature of a response is an assertion. A formatting violation, for example, can be evaluated directly against a task requirement without being treated as a representational claim. Such a violation falls outside Equation (5), although it remains relevant to task-obligation fidelity. Having established the governing reference and the assertion-reference relation, we can now distinguish the error classifications that may apply.

D. Hallucination and Task-Obligation Fidelity

The framework applies two error classifications to a delivered response. Hallucination concerns the representational content of an assertion. Task-obligation fidelity error concerns failure of the response to conform to an applicable requirement. Because a response may both misrepresent a task requirement and fail to satisfy that requirement, the two classifications may apply separately or together.

More specifically, a representational assertion may concern an external fact, a designated source, an applicable instruction, an authoritative contextual record, or an authorized project state. We classify the assertion as a hallucination if it is contradicted by, or lacks support required from, its governing reference. Hallucination is therefore a property of a delivered claim assessed after the fact relative to its governing reference. The classification does not, however, establish how the claim was produced.

A task-obligation fidelity error, by comparison, occurs when the delivered response departs from a requirement established by a governing source, instruction, contextual record, or authorized project state. The requirement may govern the form of the response, such as its word limit, or require faithful representation of a source.

Consider, for example, a request for a source-faithful summary of no more than 200 words. An accurate summary of 300 words violates the length requirement and therefore contains a task-obligation fidelity error; the excess length does not, by itself, constitute hallucination. If the response also states that it complies with the 200-word limit, that statement is a representational assertion contradicted by the instruction. In the latter scenario, the response would therefore exhibit both hallucination and task-obligation fidelity error.

Allowing the two error classifications to overlap avoids two distortions. If every task failure were called hallucination, a formatting violation and a fabricated citation would be placed in the same category. If false claims about instructions, contextual records, or project state were excluded from hallucination merely because they concern different kinds of governing references, representational claims with the

same relation to their governing references would be classified differently. The relevant question for hallucination classification is whether the response makes a representational claim and, if so, how that claim relates to its governing reference.

These error classifications establish what occurred, not why it occurred. A hallucination may be identified even when no production record reveals how the claim arose. Conversely, a retrieval or tool failure is not itself a hallucination if the system abstains and makes no discrepant assertion. This separation allows possible production causes to be investigated without placing an assumed cause inside the definition of the error.

E. Localization and Causal Attribution

Once a response error has been identified and classified, evaluation may turn to the production of the error. Two questions then arise: Where could the error have arisen, and what evidence shows that a condition at that location contributed to it? We use *localization* for the first inquiry—the identification of a plausible production locus or set of loci—and *causal attribution* for the second. Localization identifies where a contributing condition may have occurred, but localization alone does not establish that the condition caused or contributed to the delivered error. Causal attribution requires additional evidence of that contribution. Prior research identifies multiple mechanisms through which response error, including hallucination, may arise ([10]). Building on that research, a recent paper by this author groups plausible sources of hallucination into three broad mechanism families ([11]).

The first family, Model Approximation, concerns susceptibilities created when a finite predictive model is formed from bounded information under design-contingent choices. The second, Autoregressive Realization, concerns conditional prediction, decoding, and the dependence of later response generation on earlier selections. The third, Deployment Mediation, concerns retrieval, context construction, interaction history, tools, and post-generation processes. These families identify principal loci of susceptibility. The families may interact, and the contribution of any one family cannot be inferred from the final error alone.

To illustrate this point, consider a hallucination about a designated source. Suppose the delivered response makes a claim that the designated source contradicts or does not support. The error may have arisen because the relevant source passage was not retrieved. The passage may instead have been retrieved but omitted during context construction. Even if the passage entered the operative context, the generation process may have failed to use it. Finally, the candidate response may have represented the source accurately, but a post-generation process may have altered the claim before delivery. These production paths can result in the same delivered hallucination, although each path places the relevant susceptibility at a different stage.

Another example may be a hallucination involving an outdated factual answer. Suppose the delivered response states a fact that was once accurate but is no longer current. The outdated claim may reflect the fact that the model was trained without information about later events. It may also result from a failure to retrieve or include current information in the operative context. Even when current information is included, the generation process may realize a strong but outdated association instead. More than one of these conditions may also contribute to the same claim. The outdated answer establishes the error relative to the current governing reference, but the answer alone does not establish which condition produced it.

Distinguishing among these plausible production paths requires evidence from the production process. The available evidence determines how narrowly an evaluator can localize the error and whether the evaluator can proceed from localization to causal attribution. Retrieval and context-construction logs may show whether relevant material entered C_t . A stored candidate response may show whether post-generation processing changed the claim. Repeated trials may indicate whether the behavior is systematic or sensitive to realization, while controlled interventions may test whether changing a production condition changes the error. If these records are unavailable, as they often are in closed systems, the evaluator may identify several plausible loci but remain unable to attribute the error to any one of them.

Table 2 brings these distinctions together. It shows why classification of an error does not identify its cause. Different production paths may produce the same observed error. Conversely, failure at the same production stage may lead to different outcomes, including abstention, omission, or fabrication. A delivered response and its governing reference may therefore establish that a hallucination occurred even when the available production evidence does not establish how the hallucination arose.

Table 2. Error occurrence, possible production locus, and evidence for causal attribution

Observed condition	Response-level classification	Possible production locus and evidence needed
Retrieval returns no relevant material and the system abstains	No assertion discrepancy; availability failure	Deployment Mediation; retrieval logs establish the failure
Retrieval fails and the response invents a source-backed answer	Hallucination involving a designated source	Deployment Mediation and Autoregressive Realization may interact; retrieval, context, and generation records are needed
Accurate source material is present in the operative context, but the response contradicts it	Hallucination involving a designated source	Model Approximation or Autoregressive Realization; controlled generation or model comparisons may distinguish them
A current factual question is answered from outdated learned information	Factual hallucination if contradicted by the current reference	Model Approximation, possibly interacting with the absence of current information and later propagation
A post-generation control inserts an unsupported explanation	Possible factual or contextual hallucination	Deployment Mediation; candidate and delivered responses establish the transformation

The separation between error classification and causal attribution also affects the selection of controls. If current information needed for the task never entered the operative context, changing the decoding process alone cannot supply that information. If accurate source material entered the operative context but the generation process repeatedly failed to use it, improving retrieval may have little effect. Control selection should therefore respond to what the available production evidence establishes rather than to the model name or error classification alone. When the available production evidence does not permit a response error to be localized precisely, control selection should reflect that uncertainty.

F. Extensions of the Framework

The analysis so far concerns one delivered response produced under a given set of conditions. In practice, both governing references and production conditions may change across time or across system configurations. Time, continuing-project state, and retrieval-augmented generation illustrate how the authority-first sequence extends to these settings without changing the definitions of the two error classifications.

F.1 Time, Project State, and Propagation

Although evaluation occurs at a particular interaction time, the time governing an assertion need not be that interaction time. Instead, the task specifies or implies the temporal target to which the assertion should refer. A task may ask for a current officeholder, a historical stock price, or the rule in force on an earlier date. In some cases, the temporal target coincides with the time at which the task is issued; in other cases, it does not. A response accurate at one time may therefore be classified as a hallucination if it is contradicted by, or lacks required support from, the governing reference applicable to the task’s temporal target.

This temporal distinction also limits what can be inferred from an error. Suppose a model was trained using information available through February. On May 31, a user asks the model in a deployed system for the current price of a stock. The system supplies a price even though it has no access to market information produced after February. If that price differs from the verified May 31 price, the claim may be classified as a hallucination relative to the governing reference. But the classification does not mean that the trained model failed to learn the May 31 price, because that price did not exist in the information used for training. Nor does the classification alone establish why the deployed system supplied the incorrect price.

Project state is another case where the governing reference may change over time. Consider a continuing project in which a team initially discusses one proposal but later rejects it and approves a different course of action. Records of both proposals may remain accessible to the system. If the system is subsequently asked to continue the project, the approved decision—not the superseded proposal—forms part of the authorized project state. A response that continues the project according to the superseded proposal would therefore contain a task-obligation fidelity error. If the response also states that the superseded proposal remains approved, that assertion about the project state would be a hallucination as well. Evaluation of responses in continuing projects should therefore record not only the interaction time and the time relevant to the task, but also the authorized project state as it is updated over time.

A third temporal issue arises when an earlier error is carried into later work. Suppose a project has a completion deadline of Friday, but a system misstates the deadline as Monday and maintains the incorrect date in its project record. If the system later retrieves that statement and uses Monday to prepare a work schedule, the earlier error will propagate into the later response. Whether an evaluator can identify this propagation, however, depends on the available production evidence. Relevant evidence may include a record of the initial error, a record showing that the incorrect date was retained and retrieved, and evidence that the system used that date in producing the later response.

F.2 Retrieval and Context Construction

Time and project state discussed above show how governing references and production conditions may change across interactions. Retrieval introduces a different issue within response production: information available to a system may differ from the information ultimately supplied to the model. Retrieval-augmented generation, or RAG, is a deployment process through which a system retrieves external information and places selected material in the operative context ([12]). RAG is commonly used to provide current, private, or task-specific information. Its production stages make visible the distinctions among information available to the system, information placed in the operative context, and the reference governing evaluation.

When a system receives a task, a typical RAG process constructs a query, uses the query to search an available corpus, ranks the retrieved material, and places selected material in the operative context. Each stage creates a point at which relevant information can be omitted, altered, or assigned the wrong priority. The query may misstate the information need, the search may omit relevant material, or the ranking process may favor an inapplicable source. During context construction, relevant content may be omitted, or the selected material may lose its provenance, temporal qualifications, or authority labels. Prior research documents these RAG failure points ([13], [14]).

Because the stages are sequential, adequate performance at an earlier stage does not establish adequate performance at a later stage. A search may retrieve a relevant document, for example, but the passage needed for the task may still be omitted from the operative context. Even if the passage enters the operative context, the generation process may disregard it and rely instead on conflicting information learned during training. Retrieval may also impair the response if the material supplied to the model is outdated, inapplicable, or unauthorized. The information used in the delivered response therefore depends not only on what the system retrieves, but also on what it selects, preserves during context construction, and uses during generation.

These production questions are separate from the question of evaluative authority. The fact that material was retrieved or placed in the operative context does not make that material the governing reference. Retrieved material may itself be the governing reference, may conflict with that reference, or may have no bearing on the assertion being evaluated. The evaluator still needs to identify the reference governing each assertion and classify the relation between the assertion and that reference. RAG therefore only changes how information enters response production and where an error may arise; it does not change how assertions in the delivered response are evaluated.

G. Discussion and Conclusion

G.1 Consequence and controls

Occurrence of an error does not determine its consequence. The consequence of a hallucination or task-obligation fidelity error depends on the task, the reliance placed on the response, and whether the error is retained or propagated. Consider the same false statement in two settings. In a casual exchange in which no one relies on the statement, the error may have little effect. If the statement is instead entered into a research record, compliance analysis, or organizational decision process, later work may rely on it and extend its effects.

Consequence assessment helps determine the significance and urgency of a governance response, but it does not by itself identify the appropriate control. Control selection should also reflect what the available production evidence establishes about how the error arose. If information needed for the task was unavailable to the system, the response may require better information sources or access to current data. If the information was available but did not enter the operative context, retrieval or context construction may require revision. If the correct information entered the operative context but was not used in the response, verification or changes to the generation process may be appropriate. If repeated trials under the same production conditions yield different results, changes to decoding or verification may reduce that variation. If a post-generation process introduced the error, the rules or procedures governing that process may need revision. When causal attribution remains unresolved, improved production records or controlled testing may be the appropriate first response.

These examples also show why controls are not interchangeable. Retrieval cannot correct an error when the retrieved source is itself outdated or inapplicable. Changing the decoding process cannot supply information that never entered the operative context. A more capable model may still answer a current-information question incorrectly if the deployed system provides no current information. Evaluation and error management should therefore concern a configured system performing a specified task rather than a model name considered in isolation.

G.2 Evaluation sequence

Our analysis can now be brought together as an eight-step sequence for evaluating a configured system that performs a specified task:

1. Specify the task and the applicable authority policy.
2. Identify the assessable assertions and applicable task obligations.
3. Select the governing reference for each assertion.
4. Classify each assertion-reference relation as supported, contradicted, unsupported, or unresolved.
5. Determine whether the response contains hallucination, task-obligation fidelity error, or both.
6. Identify plausible production loci and make causal claims only to the extent supported by production evidence.
7. Assess the consequences of each error, including possible reliance and propagation.
8. Select controls appropriate to the supported findings and the deployment context.

These steps follow the analytical order developed in this paper. The sequence begins with task specification and the applicable authority policy because they provide the basis for selecting governing references and evaluating the response. An alternative beginning with an assumed cause may distort both error classification and control selection.

G.3 Limitations

Our framework has several important limitations. First, the framework requires an authority policy but does not supply one by itself. The governing references selected and the resulting error classifications are therefore conditional on the authority policy provided for the task. Although the framework shows how different authority policies may lead to different evaluation results, it cannot determine which policy an evaluator or organization should adopt.

Second, the framework has no empirical component. It does not estimate the frequency of hallucination or task-obligation fidelity error, or determine the relative contribution of different production mechanisms to errors, or prescribe error controls independently of the organizational context. Nor does it cover the specifics of adverse implications or effects from the identified and classified errors. Empirical investigations into these components will need to be carried out in the future.

Third, error localization and causal attribution depend on the availability and completeness of production evidence. When the production evidence is not available or complete, precise localization and causal attribution of the response error may not be possible.

G.4 Conclusion

A generative AI response delivered for a task is produced by a configured system, whereas the assertions in that response are evaluated relative to governing references selected for the task. Response production and response evaluation therefore address different questions. Our framework addresses this distinction through an authority-first evaluation sequence. The sequence begins by specifying the task and authority policy, identifying the assessable assertions, selecting the governing reference for each assertion, and classifying the relation between the assertion and that reference. It then determines whether the response contains hallucination, task-obligation fidelity error, or both. Finally, it separates error classification from error localization and causal attribution and treats the consequences of an error as a further question.

The authority-first sequence provides a common structure for evaluating external factual claims, designated-source tasks, continuing-project work, and responses produced with RAG, as well as for selecting system controls. Whether the framework improves evaluator agreement or leads to better control choices remains an empirical question.

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